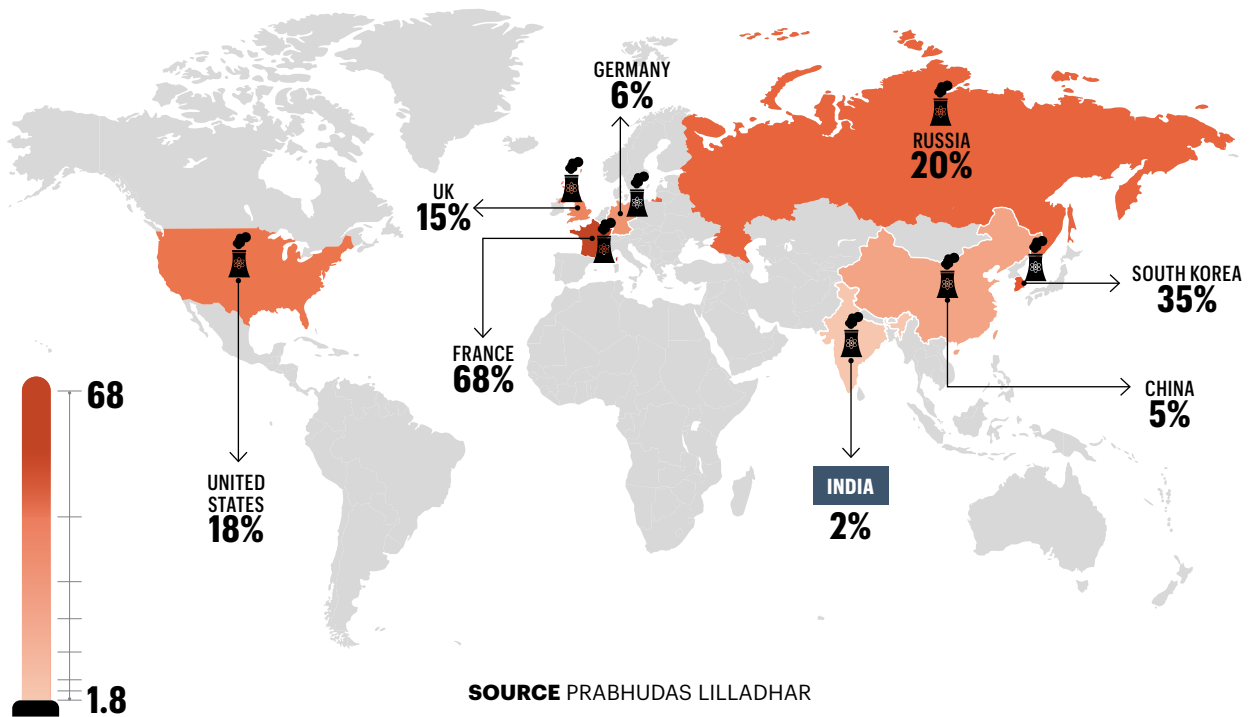


INDIA LAGS

France has the highest share of nuclear in its energy mix, at 68%. South Korea is second at 35%



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Commission, Anil Kakodkar, say India cannot achieve the target of 100GW nuclear capacity by 2047 by following the three-stage nuclear programme. He believes the country must look for different nuclear technologies under various stages of development.

“You don’t have to go through in that sequence... as stage one, stage two. It happened well and good, but if not [completed], I

think nothing is lost. Now, we should move to thorium, accumulate that and accelerate stage three,” he tells *Business Today*. His idea is to take a slightly altered path to a three-stage programme which can advance the thorium usage for India.

The Kalpakkam FBR’s achievement has put the spotlight on utilising the vast thorium resources in India. Stage 3 reac-

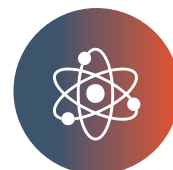
tors require an initial loading of Uranium 233, a material that does not occur in nature and must be bred from thorium by bombarding it with neutrons.

“Under the classical pathway, the only source of those neutrons at scale is the FBR fleet. But FBRs take decades to build, and their early fuel is precious Pu-239 from Stage 1—itself in limited supply. It is a slow com-

THE AIM IS TO GENERATE NEARLY 5%, OR 100 GW, BY 2047



EIGHT NUCLEAR REACTORS OF 6.6 GW CAPACITY UNDER CONSTRUCTION



INDIA HAS 25% OF THE WORLD’S THORIUM RESERVES